Unit 2: Output Primitives (6 Hrs.)

- Scan conversions of point, line, circle and ellipse:
- DDA algorithm and Bresenham algorithm for line drawing, midpoint algorithm for circle, midpoint algorithm for ellipse drawing (Mathematical derivation for above algorithms is expected),
- Filled Area Primitive: Scan linePolygon Fill algorithm, inside outside tests, Boundary Fill and Flood fill algorithm.

❑ Scan Conversion Definition

- It is a process of representing graphics objects a collection of pixels. The graphics objects are continuous. The pixels used are discrete. Each pixel can have either on or off state.

- The circuitry of the video display device of the computer is capable of converting binary values (0, 1) into a pixel on and pixel off information. 0 is represented by pixel off. 1 is represented using pixel on. Using this ability graphics computer represent picture having discrete dots.

- Any model of graphics can be reproduced with a dense matrix of dots or points. Most human beings think graphics objects as points, lines, circles, ellipses. For generating graphical object, many algorithms have been developed.

❑ Advantage of developing algorithms for scan conversion

- Algorithms can generate graphics objects at a faster rate.
- Using algorithms memory can be used efficiently.
- Algorithms can develop a higher level of graphical objects.

Examples of objects which can be scan converted

- Point
- Line
- Sector
- Arc
- Ellipse
- Rectangle
- Polygon
- Characters
- Filled Regions

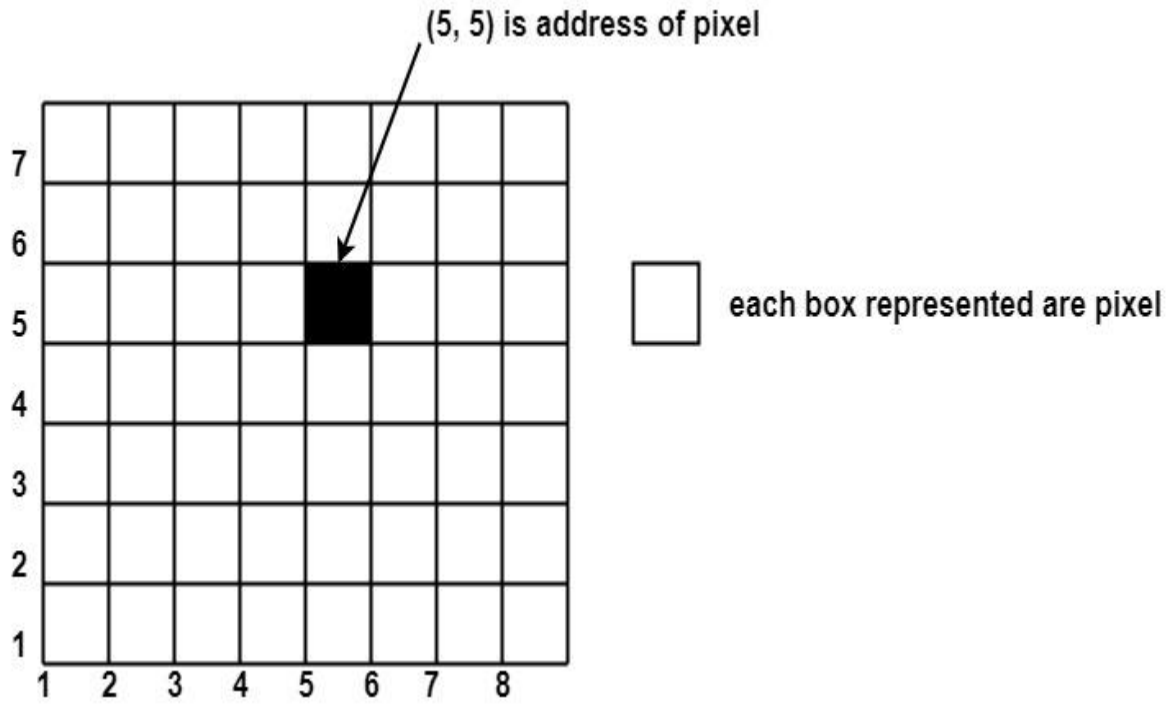
The process of converting is also called as rasterization. The algorithms implementation varies from one computer system to another computer system. Some algorithms are implemented using the software. Some are performed using hardware or firmware. Some are performed using various combinations of hardware, firmware, and software.

Pixel or Pel:

The term pixel is a short form of the picture element. It is also called a point or dot. It is the smallest picture unit accepted by display devices. A picture is constructed from hundreds of such pixels. Pixels are generated using commands. Lines, circle, arcs, characters; curves are drawn with closely spaced pixels. To display the digit or letter matrix of pixels is used.

The closer the dots or pixels are, the better will be the quality of picture. Closer the dots are, crisper will be the picture. Picture will not appear jagged and unclear if pixels are closely spaced. So the quality of the picture is directly proportional to the density of pixels on the screen.

Pixels are also defined as the smallest addressable unit or element of the screen. Each pixel can be assigned an address as shown in fig:

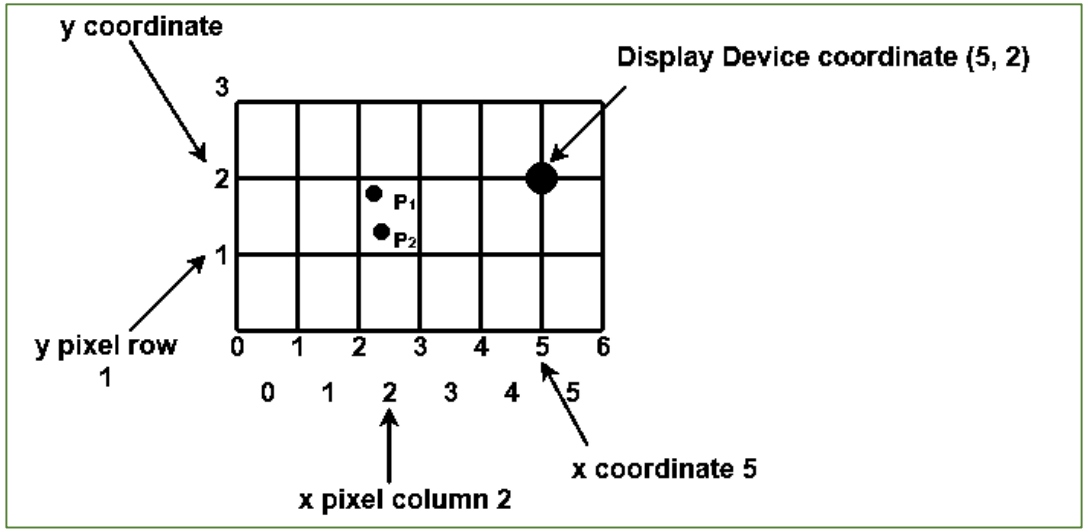


P (5, 5) used to represent a pixel in the 5th row and the 5th column. Each pixel has some intensity value which is represented in memory of computer called a frame buffer. Frame Buffer is also called a refresh buffer. This memory is a storage area for storing pixels values using which pictures are displayed. It is also called as digital memory. Inside the buffer, image is stored as a pattern of binary digits either 0 or 1. So there is an array of 0 or 1 used to represent the picture. In black and white monitors, black pixels are represented using 1's and white pixels are represented using 0's. In case of systems having one bit per pixel frame buffer is called a bitmap. In systems with multiple bits per pixel it is called a pixmap.

❑ Scan Converting a Point

- Each pixel on the graphics display does not represent a mathematical point. Instead, it means a region which theoretically can contain an infinite number of points. Scan-Converting a point involves illuminating the pixel that contains the point.

Example: Display coordinates points $P_1\ (2\frac{1}{4}, 1\frac{3}{4})$& $P_2(2\frac{2}{3}, 1\frac{1}{4})$ as shown in fig would both be represented by pixel (2, 1). In general, a point p (x, y) is represented by the integer part of x & the integer part of y that is pixels [(INT (x), INT (y).



❑DDA algorithm

- DDA stands for Digital Differential Analyzer. It is an incremental method of scan conversion of line. In this method calculation is performed at each step but by using results of previous steps.

Step1: Start Algorithm

Step2: Declare x1,y1,x2,y2,dx,dy,x,y as integer variables.

Step3: Enter value of x1,y1,x2,y2.

Step4: Calculate dx = x2-x1 and dy = y2-y1

m = dy/dx =(y2-y1)/(x2-x1)

dy = m.dx , dx = dy/m

Step6: check If (dx) > = dy

Assign dx =1 and find

$x_{i+1} = x_i + dx$ **and**

$y_{i+1} = y_{1+}dy = y_{1+}dx.m = y_1 + m$ **[dx=1]**

Repeat to find x and y till to end point

Step7: If (dx) < dy

Assign dy =1 and find

$x_{i+1} = x_i + dx$

$= x_i + dy/m = x_1 + 1/m$

and

$y_{i+1} = y_{1+}dy = y_1 + 1$ [dy =1]

Repeat to find x and y till to end point

Example-1 DDA Algorithm

Draw a line from (1,1) to (8,7) using DDA a

Step 1 x1=1,y1=1 and x2=8,y2=7

Step 2

dy = 6 , dx = 7

m = 6/7 = 0.85

Step comparing dx and dy

dx>dy 7 > 6 True hence

assign dx = 1

and we need to find

xi+1 = x1+dx = 1+1 =2

yi+2 =y1+dy = 1+ dx.m = 1+1x 0.85 = 1+0.85 =1.85

Repeating for next x and y value

Xi+1 =x1+dx = 2 +1 =3

yi+1 = y1+dy = 1.85 +dx.m =1.85+1x 0.85

=1.85+0.85=2.7

Like wise till 8,7

X1	Y1	Xi+1	Yi+1
1	1	2	1.85 ~2
2	1.85	3	2.7 ~3
?	?	?	?
?	?	?	?
?	?	?	?

Example -2 If a line is drawn from (2, 3) to (6, 15) with use of DDA. How many points will needed to generate such line?

Example: 3 If a line is drawn from (0, 0) to (10, 5) with DDA

❑ Bresenham algorithm for line drawing

Algorithm

Step 1 find

slope (m) =dy/dy = (y2-y1)/(x2-x1)

Step 2 find decision parameter

p =2dy –dx

Step3 check following condition and proceed as :

if m>=1 Then go through and check p and proceed as

NO

if m<1 Then go through and check p and proceed as

YEs

If p>=0 Then find

$X_{i+1} = x_i + 1$

$Y_{I+1} = y_1 + 1$

$P_1 = p_0 + 2dx - dy$

If p<0 Then find

$X_{i+1} = x_i$ [No Change]

$Y_{I+1} = y_1 + 1$

$P_1 = p_0 + 2dx$

If p>=0 Then find

$X_{i+1} = x_i + 1$

$Y_{I+1} = y_1 + 1$

$P_1 = p_0 + 2dy - 2dx$

If p<0 Then find

$X_{i+1} = x_i + 1$

$Y_{I+1} = y_1$ [No change]

$P_1 = p_0 + 2dy$

Example-1 Bresenham algorithm for line drawing A(1,1) and (5,3)

Solution :-

Step 1

Slope (m) = (3-1) / (5-1) = 2/4 = 0.5

Step 2

P =2x2-4 = 0

Step 3

Here m<1 and p =0 since we need to find

If p>=0 Then find

$X_{i+1} = x_i + 1 \quad = 1 + 1 = 2$

$Y_{I+1} = y_1 + 1 \quad = 1 + 1 = 2$

$P_1 = p_0 + 2dx - 2dy \quad = 0 + 2x2 - 2x4 = -4$

Now again we have P1<0 since next step we need to find

$X_{i+1} = x_i + 1 \quad = 2 + 1 = 3$

$Y_{I+1} = y_1$ [No change] = 2

$P_2 = p_0 + 2dx \quad = -4 + 2\ x2 = 0$

Now again we have p2 = 0 since next step we need to find

Class work

P	Xi	Yi	Xi+1	Yi+1	
0	1	1	2	2	
-4	2	2	3	2	
0	3	2	4	3	
-4	4	3	?	?	?

Complete in class

Repeat the process till (5,3)

Example: Starting and Ending position of the line are (1, 1) and (8, 5).

Example-2 Digitize a line with end points (20,10) and (30,18) using BSA

1. slope m = dy/dx = 8/10 = 0.8

2. Decision Parameter p_0 = 2dy –dx = 6

3. since m<1 and p >6 since we follow the

If p>=0 Then find

$X_{i+1} = x_i + 1$

$Y_{I+1} = y_1 + 1$

$P_1 = p_0 + 2dy - 2dx$

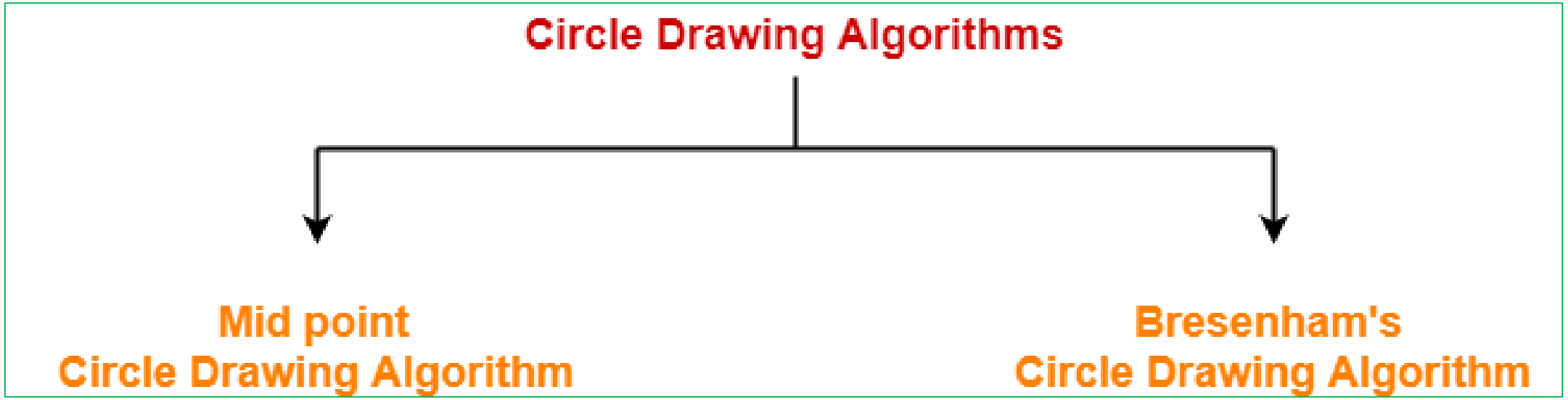
Complete all :- Class work

P	Xi	Yi	Xi+1	Yi+1	
0	20	10	21	11	
6	?	?	?	?	?
2	?	?	?	?	?
?	?	?	?	?	?

Example-3 Digitize a line with end points (10,15) and (15,18) using BSA

❑ Differentiate between DDA Algorithm and Bresenham's Line Algorithm:

DDA Algorithm	Bresenham's Line Algorithm
1. DDA Algorithm use floating point, i.e., Real Arithmetic.	1. Bresenham's Line Algorithm use fixed point, i.e., Integer Arithmetic
2. DDA Algorithms uses multiplication & division its operation	2.Bresenham's Line Algorithm uses only subtraction and addition its operation
3. DDA Algorithm is slowly than Bresenham's Line Algorithm in line drawing because it uses real arithmetic (Floating Point operation)	3. Bresenham's Algorithm is faster than DDA Algorithm in line because it involves only addition & subtraction in its calculation and uses only integer arithmetic.
4. DDA Algorithm is not accurate and efficient as Bresenham's Line Algorithm.	4. Bresenham's Line Algorithm is more accurate and efficient at DDA Algorithm.
5.DDA Algorithm can draw circle and curves but are not accurate as Bresenham's Line Algorithm	5. Bresenham's Line Algorithm can draw circle and curves with more accurate than DDA Algorithm.



- Bresenham Circle Drawing Algorithm-

  - Given the center point and radius of circle,
  - Bresenham Circle Drawing Algorithm attempts to generate the points of one octant.

Procedure-

Given-

- Centre point of Circle = (X_0, Y_0)
- Radius of Circle = R

➢ The points generation using Bresenham Circle Drawing Algorithm involves the following steps-

Step-01: [Case -1 Center (0,0)]

Assign the starting point coordinates (X_0, Y_0) as-

$X_0 = 0$
$Y_0 = R$

Step-02:

Calculate the value of initial decision parameter P0 as-

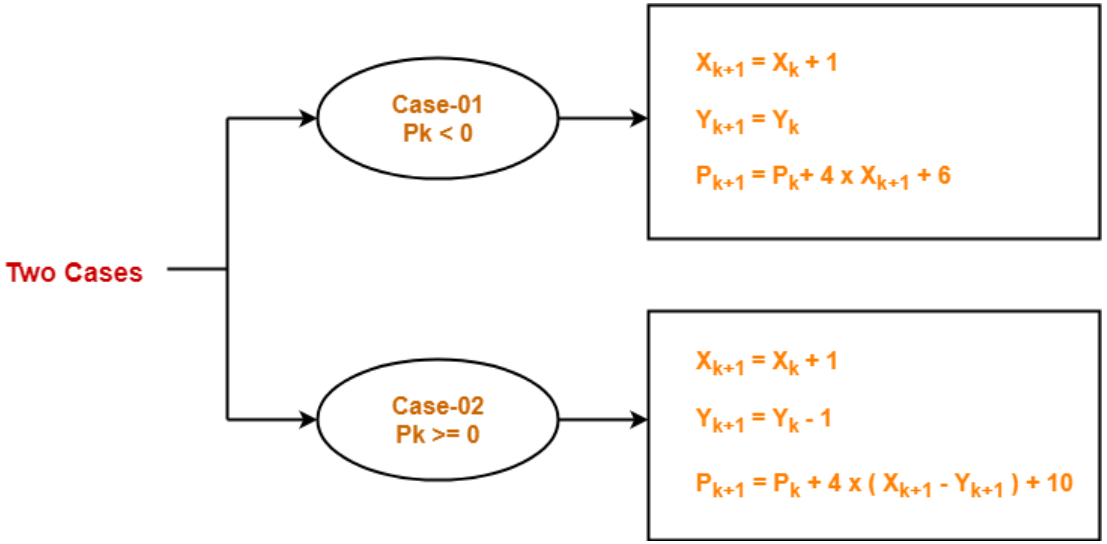
$P_0 = 3 - 2 \times R$

Step-03:

➢ Suppose the current point is (X_k, Y_k) and the next point is (X_{k+1}, Y_{k+1}).

➢ Find the next point of the first octant depending on the value of decision parameter P_k.

Follow the below two cases-

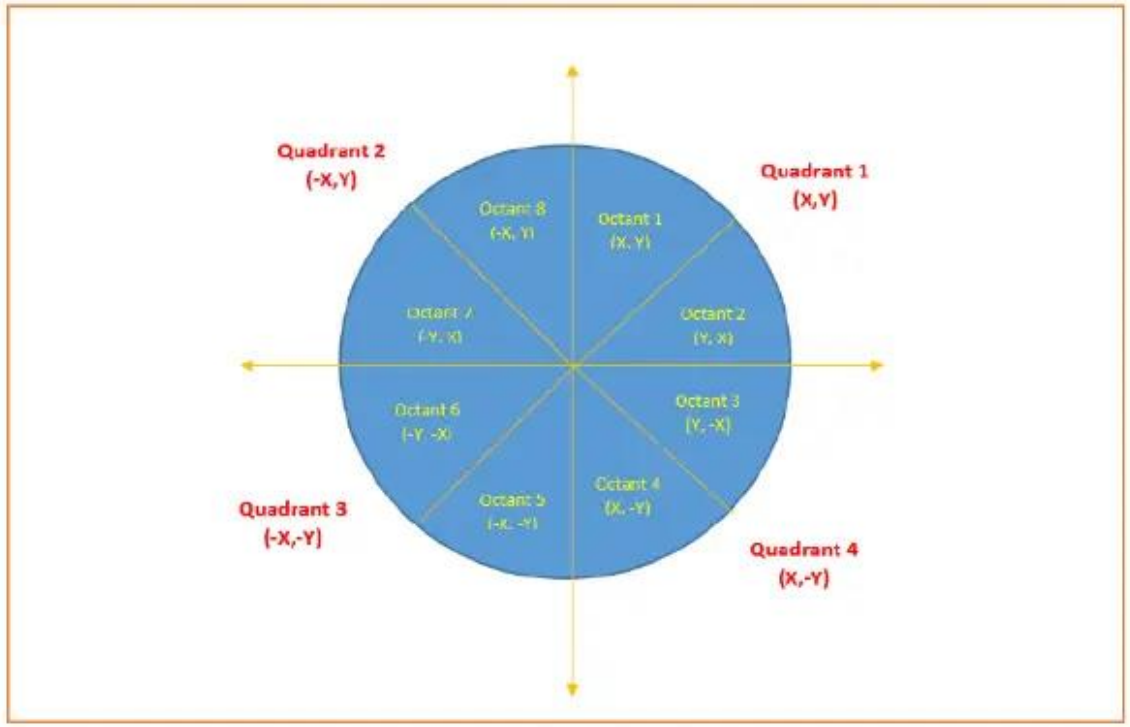


Step-04: [CASE 2]

- If the given centre point (X_0, Y_0) is not (0, 0), then do the following and plot the point-

$$X_{plot} = X_c + X0$$
$$Y_{plot} = Y_c + Y0$$

Here, (X_c, Y_c) denotes the current value of X and Y coordinates.



Step-05:

Keep repeating Step-03 and Step-04 until $X_{plot} >= Y_{plot}$.

Step-06 generates all the points for one octant. [For all Case]

- To find the points for other seven octants, follow the eight symmetry property of circle.

Problem-01:

Given the center point coordinates (0, 0) and radius as 8, generate all the points to form a circle.

Solution-

Given-

Centre Coordinates of Circle $(X_0, Y_0) = (0, 0)$

Radius of Circle = 8

Step-01:

Assign the starting point coordinates (X_0, Y_0) as-

$X_0 = 0$

$Y_0 = R = 8$

Step-02:

Calculate the value of initial decision parameter P0 as-

$P_0 = 3 - 2 \times R$

$P_0 = 3 - 2 \times 8$

$P_0 = -13$

Step-03:

As $P_{initial} < 0$, so case-01 is satisfied.

Thus,

$X_{k+1} = X_{k+1} = 0 + 1 = 1$

$Y_{k+1} = Y_k = 8$

$P_{k+1} = P_k + 4 \times X_{k+1} + 6 = -13 + (4 \times 1) + 6 = -3$

Step-04:

This step is not applicable here as the given center point coordinates is (0, 0).

Step-05:

Step-03 is executed similarly until $X_{k+1} >= Y_{k+1}$ as follows-

P_k	P_{k+1}	(X_{k+1}, Y_{k+1})
		(0, 8)
-13	-3	(1, 8)
-3	11	(2, 8)
11	5	(3, 7)
5	7	(4, 6)
7		(5, 5)
Algorithm Terminates These are all points for Octant-1.		

- Now, the points of octant-2 are obtained using the mirror effect by swapping X and Y coordinates.

Octant-1 Points	Octant-2 Points
(0, 8)	(5, 5)
(1, 8)	(6, 4)
(2, 8)	(7, 3)
(3, 7)	(8, 2)
(4, 6)	(8, 1)
(5, 5)	(8, 0)
These are all points for Quadrant-1.	

- Now, the points for rest of the part are generated by following the signs of other quadrants.
- The other points can also be generated by calculating each octant separately.
- Here, all the points have been generated with respect to quadrant-1-

uadrant-1 (X,Y)	Quadrant-2 (-X,Y)	Quadrant-3 (-X,-Y)	Quadrant-4 (X,-Y)
(0, 8)	(0, 8)	(0, -8)	(0, -8)
(1, 8)	(-1, 8)	(-1, -8)	(1, -8)
(2, 8)	(-2, 8)	(-2, -8)	(2, -8)
(3, 7)	(-3, 7)	(-3, -7)	(3, -7)
(4, 6)	(-4, 6)	(-4, -6)	(4, -6)
(5, 5)	(-5, 5)	(-5, -5)	(5, -5)
(6, 4)	(-6, 4)	(-6, -4)	(6, -4)
(7, 3)	(-7, 3)	(-7, -3)	(7, -3)
(8, 2)	(-8, 2)	(-8, -2)	(8, -2)
(8, 1)	(-8, 1)	(-8, -1)	(8, -1)
(8, 0)	(-8, 0)	(-8, 0)	(8, 0)
These are all points of the Circle.			

Problem-02:
Given the center point coordinates (10, 10) and radius as 10, generate all the points to form a circle.

Solution-

Given-
Centre Coordinates of Circle $(X_0, Y_0) = (10, 10)$
Radius of Circle = 10

Step-01:
Assign the starting point coordinates (X0, Y0) as-

$X_0 = 0$
$Y_0 = R = 10$

Step-02:

Calculate the value of initial decision parameter P_0 as-
$P_0 = 3 – 2 \times R$
$P_0 = 3 – 2 \times 10$
$P_0 = -17$

Step-03:

As $P_{initial} < 0$, so case-01 is satisfied.

Thus,
- $X_{k+1} = X_k + 1 = 0 + 1 = 1$
- $Y_{k+1} = Y_k = 10$
- $P_{k+1} = P_k + 4 \times X_{k+1} + 6 = -17 + (4 \times 1) + 6 = -7$

Step-04:

This step is applicable here as the given centre point coordinates is (10, 10).

$X_{plot} = X_c + X_0 = 1 + 10 = 11$
$Y_{plot} = Y_c + Y_0 = 10 + 10 = 20$

Step-05:

Step-03 and Step-04 are executed similarly until $X_{plot} \Rightarrow Y_{plot}$ as follows-

P_k	P_{k+1}	(X_{k+1}, Y_{k+1})	(X_{plot}, Y_{plot})
		(0, 10)	(10, 20)
-17	-7	(1, 10)	(11, 20)
-7	7	(2, 10)	(12, 20)
7	-7	(3, 9)	(13, 19)
-7	15	(4, 9)	(14, 19)
15	13	(5, 8)	(15, 18)
13	19	(6, 7)	(16, 17)
Algorithm Terminates These are all points for Octant-1.			

- Algorithm calculates all the points of octant-1 and terminates.
- Now, the points of octant-2 are obtained using the mirror effect by swapping X and Y coordinates.

Octant-1 Points	Octant-2 Points
(10, 20)	(17, 16)
(11, 20)	(18, 15)
(12, 20)	(19, 14)
(13, 19)	(19, 13)
(14, 19)	(20, 12)
(15, 18)	(20, 11)
(16, 17)	(20, 10)
These are all points for Quadrant-1.	

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- Now, the points for rest of the part are generated by following the signs of other quadrants.
- The other points can also be generated by calculating each octant separately.
- Here, all the points have been generated with respect to quadrant-1-

Quadrant-1 (X,Y)	Quadrant-2 (-X,Y)	Quadrant-3 (-X,-Y)	Quadrant-4 (X,-Y)
(10, 20)	(10, 20)	(10, 0)	(10, 0)
(11, 20)	(9, 20)	(9, 0)	(11, 0)
(12, 20)	(8, 20)	(8, 0)	(12, 0)
(13, 19)	(7, 19)	(7, 1)	(13, 1)
(14, 19)	(6, 19)	(6, 1)	(14, 1)
(15, 18)	(5, 18)	(5, 2)	(15, 2)
(16, 17)	(4, 17)	(4, 3)	(16, 3)
(17, 16)	(3, 16)	(3, 4)	(17, 4)
(18, 15)	(2, 15)	(2, 5)	(18, 5)
(19, 14)	(1, 14)	(1, 6)	(19, 6)
(19, 13)	(1, 13)	(1, 7)	(19, 7)
(20, 12)	(0, 12)	(0, 8)	(20, 8)
(20, 11)	(0, 11)	(0, 9)	(20, 9)
(20, 10)	(0, 10)	(0, 10)	(20, 10)
These are all points of the Circle.			

Disadvantages of Bresenham Circle Drawing Algorithm-

The disadvantages of Bresenham Circle Drawing Algorithm are-

- Like Mid Point Algorithm, accuracy of the generating points is an issue in this algorithm.
- This algorithm suffers when used to generate complex and high graphical images.
- There is no significant enhancement with respect to performance.

Advantages of Bresenham Circle Drawing Algorithm-

The advantages of Bresenham Circle Drawing Algorithm are-

- The entire algorithm is based on the simple equation of circle $X^2 + Y^2 = R^2$.
- It is easy to implement.

Mid Point Circle Drawing Algorithm-

Given the center point and radius of circle,
Mid Point Circle Drawing Algorithm attempts to generate the points of one octant.

Procedure-

Given-

•Centre point of Circle = (X_0, Y_0)

•Radius of Circle = R

The points generation using Mid Point Circle Drawing Algorithm involves the following steps-

Step-01:

Assign the starting point coordinates (X_0, Y_0) as-

•$X_0 = 0$

•$Y_0 = R$

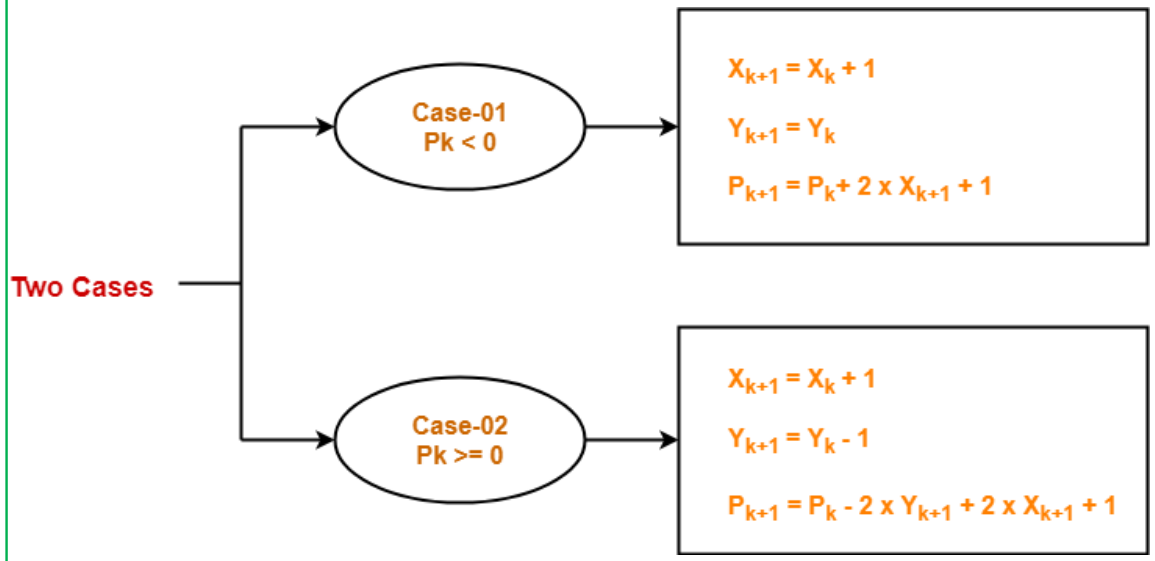
Step-02:

Calculate the value of initial decision parameter P_0 as-

$$P_0 = 1 - R$$

Step-03:

- Suppose the current point is (Xk, Yk) and the next point is (Xk+1, Yk+1).
- Find the next point of the first octant depending on the value of decision parameter Pk.
- Follow the below two cases-



Step-04:

If the given centre point (X_0, Y_0) is not (0, 0), then do the following and plot the point-

- $X_{plot} = X_c + X_0$
- $Y_{plot} = Y_c + Y_0$

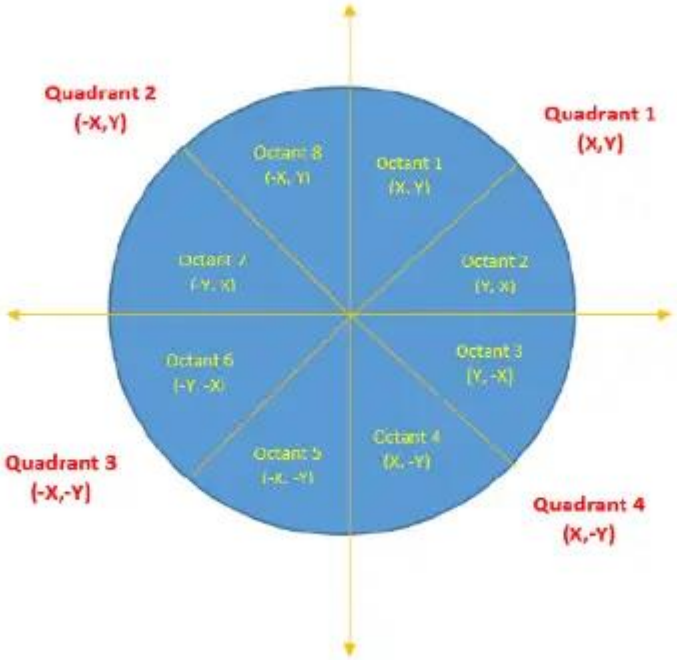
Here, (X_c, Y_c) denotes the current value of X and Y coordinates.

Step-05:

Keep repeating Step-03 and Step-04 until $X_{plot} >= Y_{plot}$.

Step-06:

Step-05 generates all the points for one octant.
To find the points for other seven octants, follow the eight symmetry property of circle.



Problem-01:

Given the centre point coordinates (0, 0) and radius as 10, generate all the points to form a circle.

Solution-

Given-

•Centre Coordinates of Circle $(X_0, Y_0) = (0, 0)$

•Radius of Circle = 10

Step-01:

Assign the starting point coordinates (X_0, Y_0) as-

•$X_0 = 0$

•$Y_0 = R = 10$

Step-02:

Calculate the value of initial decision parameter P_0 as-

$P_0 = 1 – R$

$P_0 = 1 – 10$

$P_0 = -9$

Step-03:

As $P_{initial} < 0$, so case-01 is satisfied.

Thus,

•$X_{k+1} = X_k + 1 = 0 + 1 = 1$

•$Y_{k+1} = Y_k = 10$

•$P_{k+1} = P_k + 2 \times X_{k+1} + 1 = -9 + (2 \times 1) + 1 = -6$

Step-04:

This step is not applicable here as the given centre point coordinates is (0, 0).

Step-05:

Step-03 is executed similarly until $X_{k+1} >= Y_{k+1}$ as follows-

P_k	P_{k+1}	(X_{k+1}, Y_{k+1})
		(0, 10)
-9	-6	(1, 10)
-6	-1	(2, 10)
-1	6	(3, 10)
6	-3	(4, 9)
-3	8	(5, 9)
8	5	(6, 8)
Algorithm Terminates These are all points for Octant-1.		

Octant-1 Points	Octant-2 Points
(0, 10)	(8, 6)
(1, 10)	(9, 5)
(2, 10)	(9, 4)
(3, 10)	(10, 3)
(4, 9)	(10, 2)
(5, 9)	(10, 1)
(6, 8)	(10, 0)
These are all points for Quadrant-1.	

Quadrant-1 (X,Y)	Quadrant-2 (-X,Y)	Quadrant-3 (-X,-Y)	Quadrant-4 (X,-Y)
(0, 10)	(0, 10)	(0, -10)	(0, -10)
(1, 10)	(-1, 10)	(-1, -10)	(1, -10)
(2, 10)	(-2, 10)	(-2, -10)	(2, -10)
(3, 10)	(-3, 10)	(-3, -10)	(3, -10)
(4, 9)	(-4, 9)	(-4, -9)	(4, -9)
(5, 9)	(-5, 9)	(-5, -9)	(5, -9)
(6, 8)	(-6, 8)	(-6, -8)	(6, -8)
(8, 6)	(-8, 6)	(-8, -6)	(8, -6)
(9, 5)	(-9, 5)	(-9, -5)	(9, -5)
(9, 4)	(-9, 4)	(-9, -4)	(9, -4)
(10, 3)	(-10, 3)	(-10, -3)	(10, -3)
(10, 2)	(-10, 2)	(-10, -2)	(10, -2)
(10, 1)	(-10, 1)	(-10, -1)	(10, -1)
(10, 0)	(-10, 0)	(-10, 0)	(10, 0)
These are all points of the Circle.			

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Problem-02:

Given the centre point coordinates (4, -4) and radius as 10, generate all the points to form a circle.

Solution-

Given-

•Centre Coordinates of Circle $(X_0, Y_0) = (4, -4)$

•Radius of Circle = 10

As stated in the algorithm,

•We first calculate the points assuming the centre coordinates is (0, 0).

•At the end, we translate the circle.

Step-01, Step-02 and Step-03 are already completed in Problem-01.

Now, we find the values of X_{plot} and Y_{plot} using the formula given in Step-04 of the main algorithm.

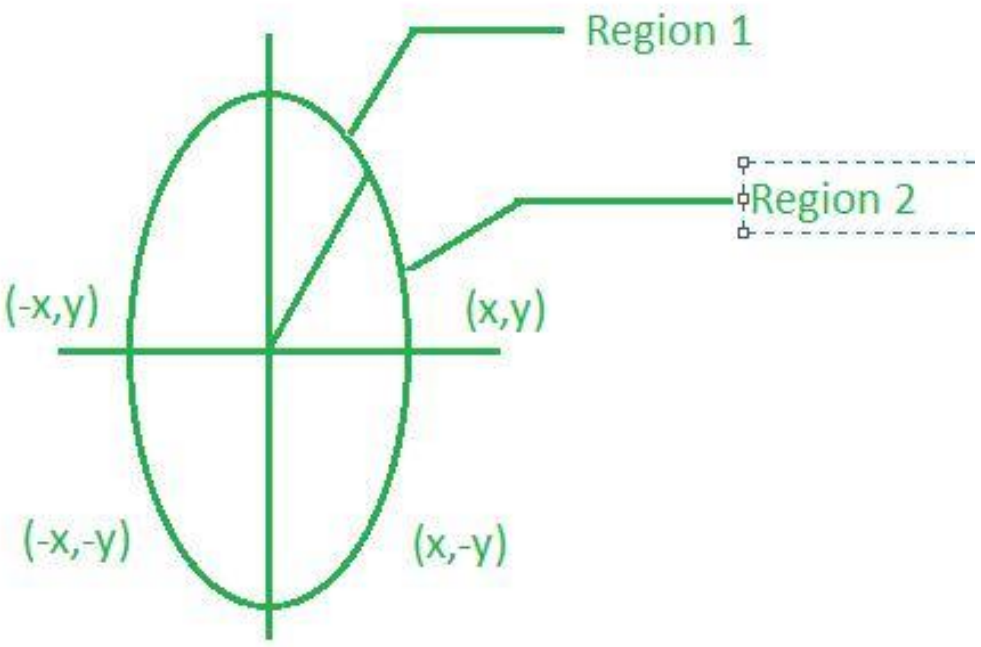
The following table shows the generation of points for Quadrant-1-

•$X_{plot} = X_c + X_0 = 4 + X_0$

•$Y_{plot} = Y_c + Y_0 = 4 + Y_0$

X_{k+1}, Y_{k+1}	(X_{plot}, Y_{plot})
(0, 10)	(4, 14)
(1, 10)	(5, 14)
(2, 10)	(6, 14)
(3, 10)	(7, 14)
(4, 9)	(8, 13)
(5, 9)	(9, 13)
(6, 8)	(10, 12)
(8, 6)	(12, 10)
(9, 5)	(13, 9)
(9, 4)	(13, 8)
(10, 3)	(14, 7)
(10, 2)	(14, 6)
(10, 1)	(14, 5)
(10, 0)	(14, 4)
These are all points for Quadrant-1	

❑ Midpoint ellipse drawing algorithm



- Midpoint ellipse algorithm plots(finds) points of an ellipse on the first quadrant by dividing the quadrant into two regions.
- Each point(x, y) is then projected into other three quadrants (-x, y), (x, -y), (-x, -y) i.e. it uses 4-way symmetry.

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$$

Suppose that $r_x = a$ and $r_y = b$ then equation will be

$$\frac{x^2}{r_x^2}+\frac{y^2}{r_y^2}=1 \quad \text{i.e}$$

$$r_y^2x^2+r_x^2y^2-r_x^2r_y^2$$

Function of ellipse:

$$\text{fellipse}(x, y)=r_y^2x^2+r_x^2y^2-r_x^2r_y^2$$

fellipse(x, y)<0 then (x, y) is inside the ellipse.

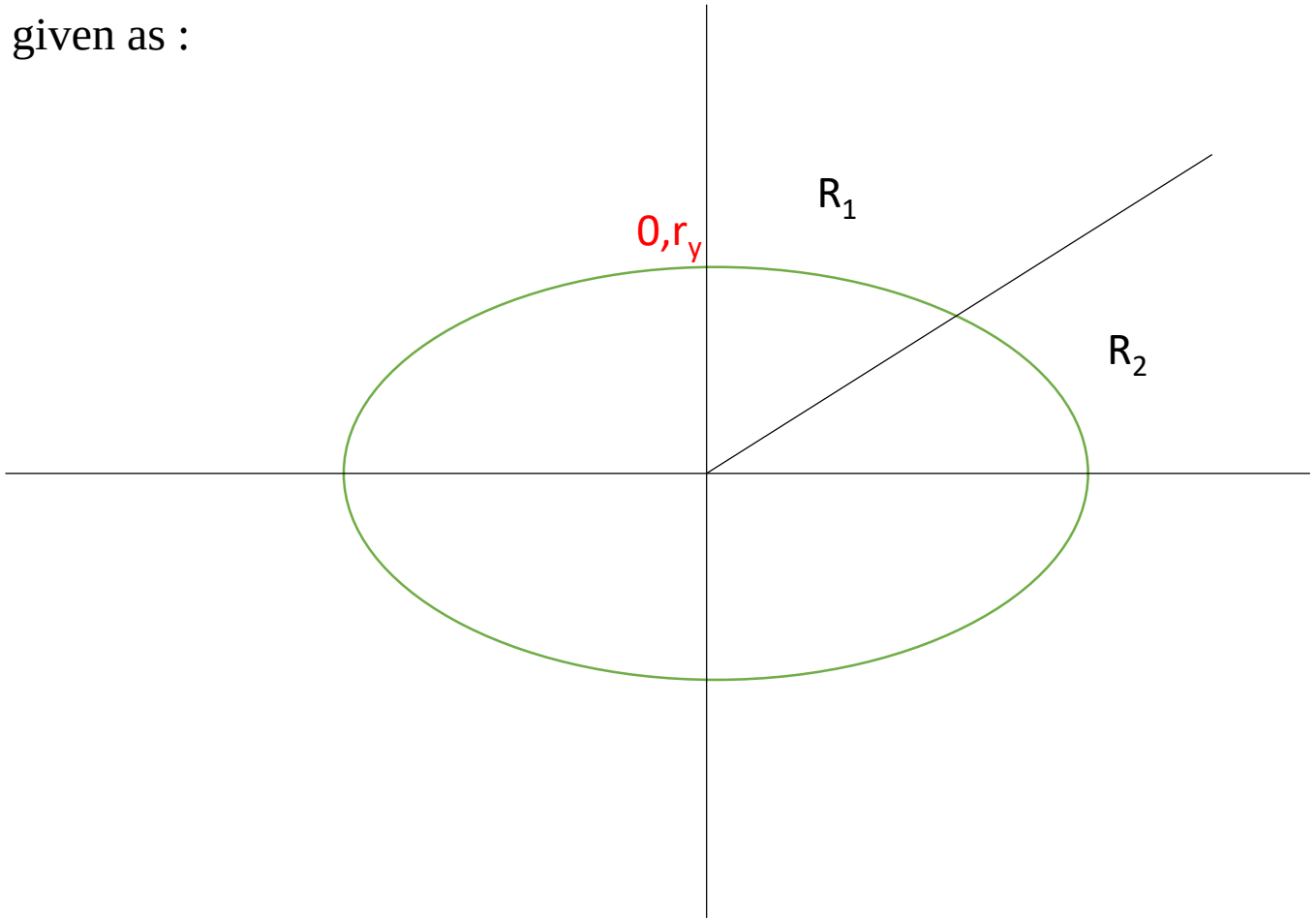
fellipse(x, y)>0 then (x, y) is outside the ellipse.

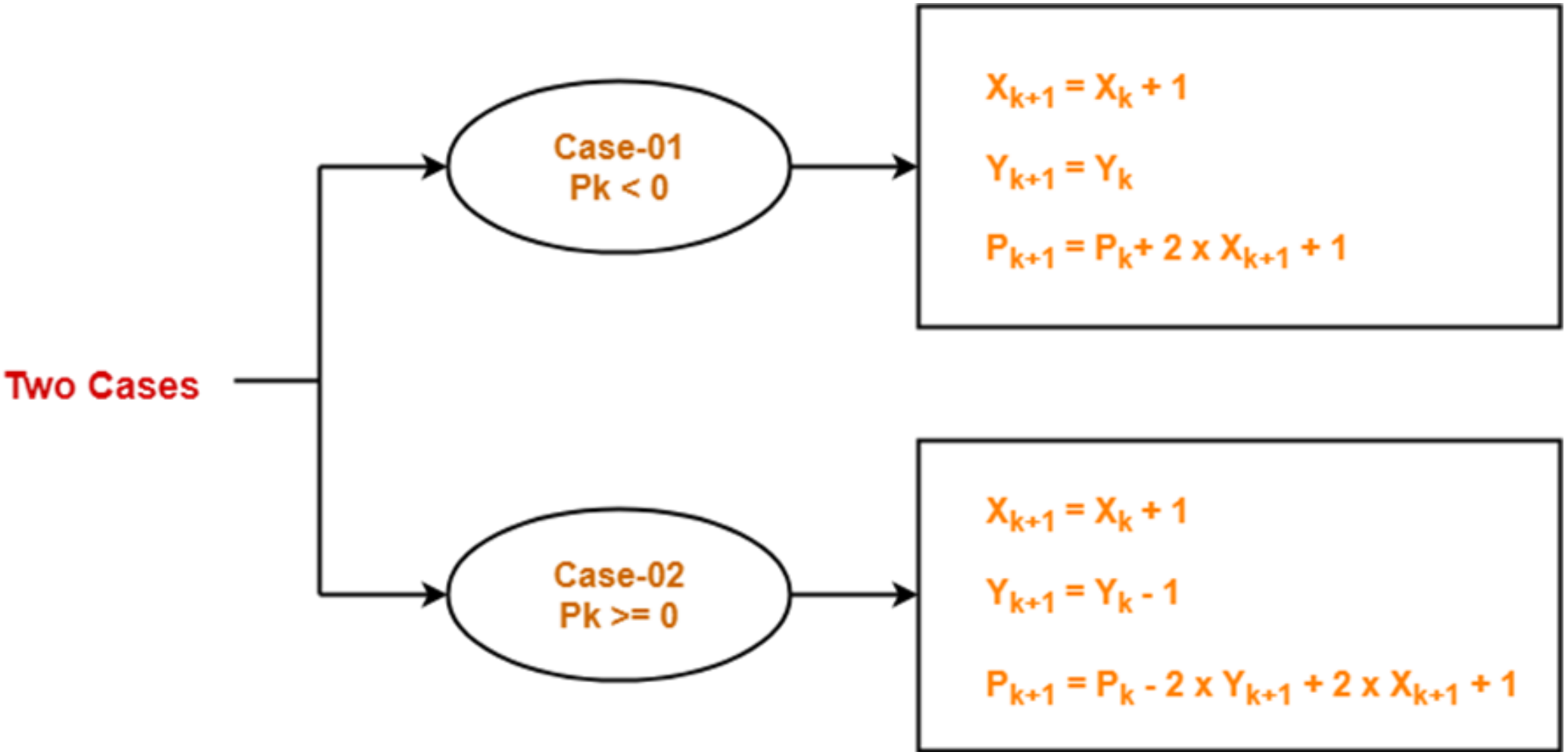
fellipse(x, y)=0 then (x, y) is on the ellipse.

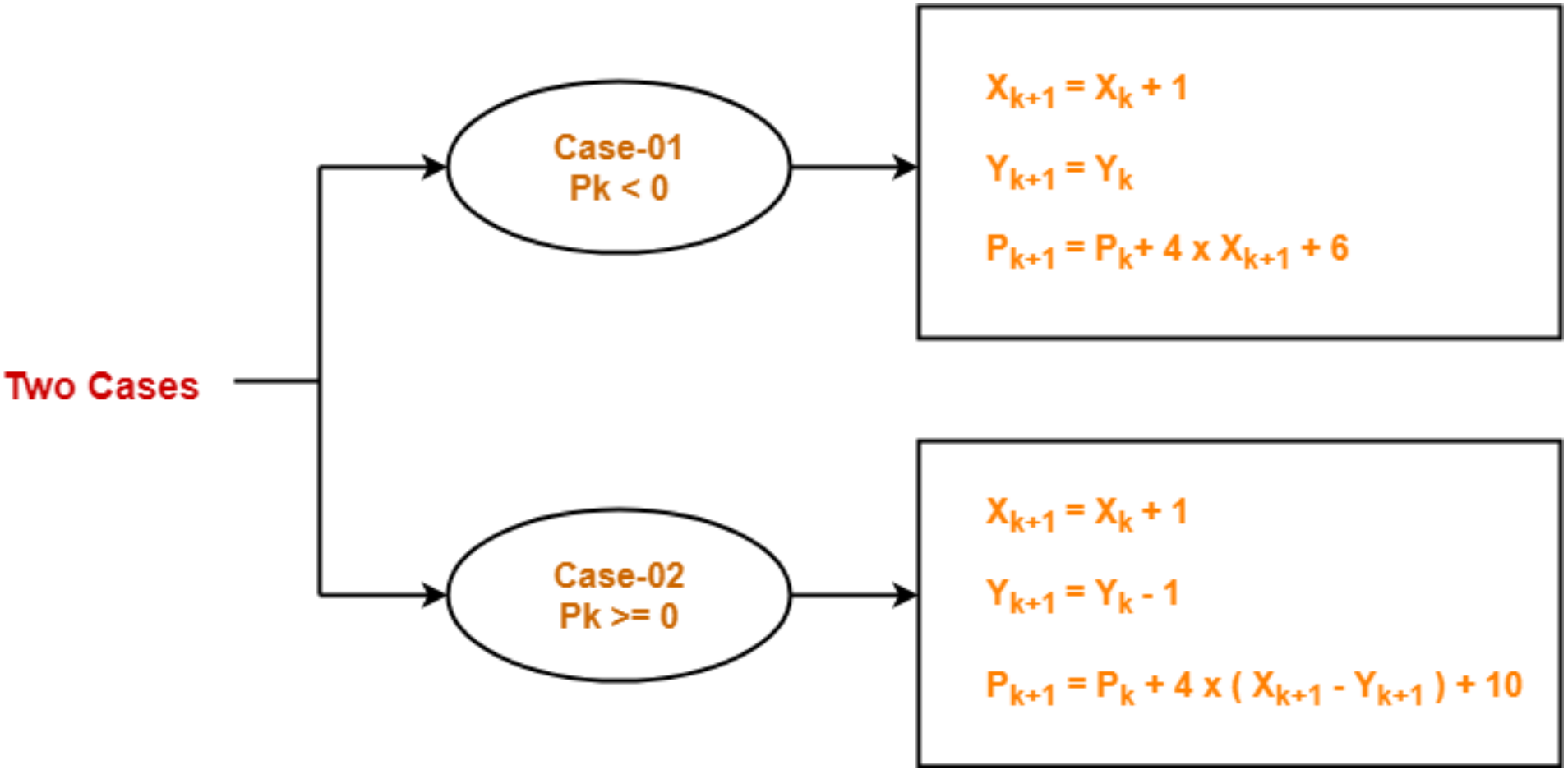
❑ Decision parameter:

- Initially, we have two decision parameters p_{10} in region 1 and **p_{20} in region 2.**
- These parameters are defined as : p_{10} in region 1 is given as :

$$p1_0 = r_y^2 + 1/4 r_x^2 - r_x^2 r_y$$




Two Cases
Case-01
Pk < 0
$X_{k+1} = X_k + 1$
$Y_{k+1} = Y_k$
$P_{k+1} = P_k + 2 \times X_{k+1} + 1$
Case-02
Pk >= 0
$X_{k+1} = X_k + 1$
$Y_{k+1} = Y_k - 1$
$P_{k+1} = P_k - 2 \times Y_{k+1} + 2 \times X_{k+1} + 1$


Two Cases
Case-01
Pk < 0
$X_{k+1} = X_k + 1$
$Y_{k+1} = Y_k$
$P_{k+1} = P_k + 4 \times X_{k+1} + 6$
Case-02
Pk >= 0
$X_{k+1} = X_k + 1$
$Y_{k+1} = Y_k - 1$
$P_{k+1} = P_k + 4 \times (X_{k+1} - Y_{k+1}) + 10$

Best of Luck